Understanding Reconstruction-Based Visual Token Compression: When and Why Does ApET Work?

Midterm Progress Report, October 2026
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1. Abstract

This project asks when and why reconstruction-based visual token compression preserves a vision-language model's answers, studying ApET [1] with LLaVA-1.5-7B [3] on GQA, MMStar and TextVQA. On a 30-question diagnostic set, reconstruction error does not consistently rank visual tokens by their measured effect on the answer, and merging is uneven: the lowest-error quarter of retained tokens absorbs two-thirds of merges. Random selection, a planned baseline, became central after small tests showed it keeping some answers ApET lost. At an average of 64 tokens, complete ApET beats random pruning by 20.9 points on TextVQA and 2.5 on GQA, with no detectable difference on MMStar. Yet relative to the uncompressed model, ApET both loses and gains answers: on MMStar at 192 tokens it loses 36 and gains 40. Which component drives these effects is not yet separated; next, we test selection and merging separately.

2. Problem Statement and Goal Recap

LLaVA-1.5 represents each image with 576 visual tokens, which dominate inference cost. ApET [1] compresses them without training: it keeps the tokens hardest to reconstruct linearly from a small basis and merges each discarded token into a similar retained one. Geometric reconstructability, however, need not coincide with preserving what the model computes downstream. The proposal asks: **RQ1:** How does reconstruction-based redundancy relate to a token's contribution to the answer, across tasks and layers? **RQ2:** How much of ApET's effectiveness comes from error-based selection, and how much from similarity-based merging? **RQ3:** Can the analysis support a lightweight improvement, such as an adaptive budget or a different basis?

First-half objectives (all met):

- Run native ApET and uncompressed LLaVA-1.5 inference in a stable, reproducible environment.
- Prepare GQA, MMStar and TextVQA evaluation subsets, baseline predictions and scoring.
- Instrument reconstruction errors, selections, merges and answer responses.

3. Methods and Implementation to Date

Reproducibility. We run the public ApET code with LLaVA-1.5-7B in FP16 on Kaggle (2×T4 GPUs), in a locked environment (Python 3.10, PyTorch 2.1.2 [7], HuggingFace Transformers 4.37.2 [8]). Our runners and scoring are separate from ApET's code; all predictions are archived, and protocols are tracked in Git.

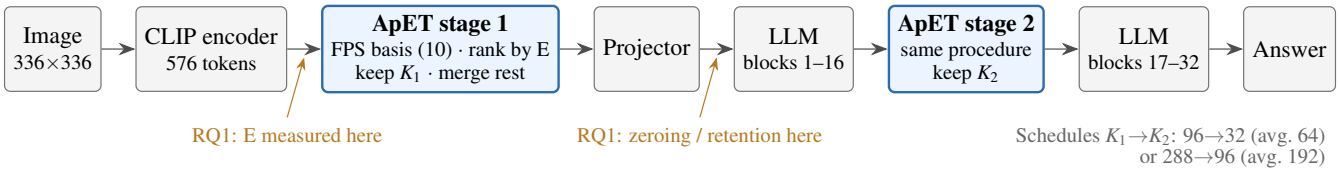


Figure 1: ApET's two compression stages, and where RQ1 (on separate uncompressed runs) measures E and answer responses.

Terms. A *schedule* $K_1 \to K_2$ keeps K_1 visual tokens after the encoder and K_2 after block 16; averaged over the 32 decoder layers, 96→32 corresponds to 64 tokens and 288→96 to 192, two of the budgets in ApET's Table 1. The *basis* is 10 tokens chosen by farthest-point sampling (FPS) in feature space; their representations are always preserved, inserted into retained token slots, though not necessarily at their original positions. *Reconstruction error* (E) measures how poorly a token is approximated by a linear combination of the basis; ApET retains the highest-E tokens. *Merging* averages each discarded token into its most cosine-similar retained token. *Random pruning* keeps the same counts at the same stages, chosen uniformly at random, with no basis, E or merging. *APR* is the share of baseline-correct answers still correct after compression; a *loss* is a correct answer turned wrong, a *gain* the reverse.

RQ1 design. For each of the 576 tokens, we measure its effect on the probability of the first token of the reference answer. *Zeroing* is the drop when only this token is set to zero; *retention*, our IH-inspired test, is the probability with

only this token kept, minus that with all visual tokens removed (a one-token adaptation of Information Horizon [2]). Both depend on the question and target answer, and neither is ground-truth importance. **RQ2 design.** We compare uncompressed inference, complete ApET and random pruning (three seeds) at both schedules, with paired 95% bootstrap intervals over images (2,000 draws). Random selection was a planned baseline; we ran this whole-procedure comparison first because early component tests showed random retention keeping some answers ApET lost, and ApET's own random ablation randomizes only the basis [1]. **Planned ablations:** single-component comparisons (Sections 4.6 and 6).

4. Preliminary Experiments and Results

Setup. RQ1 uses a 30-question diagnostic set over 24 images (12 GQA, 12 MMStar, 6 TextVQA [4–6]), all answered correctly without compression. They were chosen from our own input-only ApET runs at 288 tokens to include 11 losses and 19 preserved answers, so the set is enriched for failures and is not representative. RQ2 uses 800 GQA development questions (100 images), all 1,500 MMStar questions, and 100 TextVQA development questions. Metrics: GQA and MMStar accuracy; TextVQA ten-annotator partial-credit score; APR and loss/gain counts.

4.1 Comparison with reported ApET results

Our GQA accuracy-retention ratios (92.4%/98.6% of uncompressed accuracy at 64/192 tokens) are close to those reported for ApET (91.9%/97.3% [1]), while our TextVQA ratios are lower (83.5%/88.9% vs. 91.1%/93.5%). Our subsets, prompts and scoring may differ from the paper's, so we rely on controlled comparisons within our own setup.

4.2 Finding 1 (RQ1): reconstruction error does not consistently rank measured answer sensitivity

Across 17,280 token-level tests, median correlations between E and IH retention are -0.14 to -0.07 per dataset (11 of 30 questions positive), and between E and zeroing -0.09 to $+0.01$; individual questions range from -0.49 to $+0.40$ (Figure 2). The two measures also barely agree with each other (medians -0.03 to -0.01). Excluding padding and basis positions moves the GQA median for E versus retention to $+0.09$, without changing the overall picture. On six questions also measured after block 16, agreement did not consistently improve. E is question-independent here: in six same-image GQA pairs, both questions share one E map, yet their response maps differ (between-question correlations -0.44 to $+0.66$), though targets and no-image probabilities also differ.

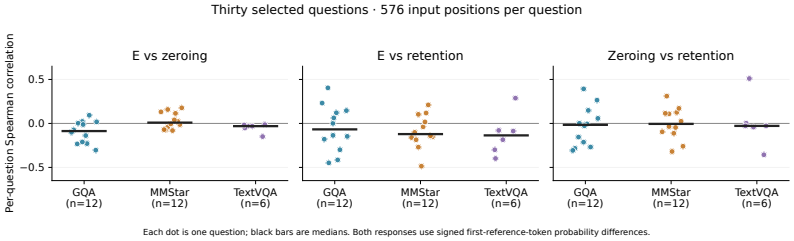


Figure 2: Per-question Spearman correlations over all 576 positions. Each dot is one question; bars are medians. No dataset shows a consistent positive relation.

4.3 Finding 2 (RQ2): merging gives retained tokens uneven roles

On the 24 diagnostic images with input-only compression to 288 tokens, 6,912 dropped tokens are merged. Ranking the 278 ordinary retained targets by E (basis slots excluded), the lowest-E quarter receives 66.1% of merges (4,569) and the highest-E quarter 1.8% (126). Dropped low-E tokens likely resemble these targets, so some retained tokens may keep distinctive features while others collect discarded information. This also explains why protecting the 28 highest-E targets from merging rerouted only 21 of 6,912 merges: that test was largely inactive. Shuffling merge assignments while keeping each target's load changed about 75% of them, yet left correctness nearly unchanged (19–20 vs. 19 of 30). Whether this division of roles helps answers is untested.

4.4 Finding 3 (RQ2): ApET's advantage over random pruning is task-dependent

At 96→32, ApET preserves 77% of TextVQA's full-credit answers versus 35% for random, while gains are similar in five of six rows (e.g., 36 vs. 35.7 on GQA). Why the advantage varies by task is not established. On MMStar, the baseline (34.7%) is near the ∼25% random-choice level reported for the benchmark [5] and predicts "A" on 70.6% of questions (28.6% of answers are A); this limits interpretation but does not establish why the ApET–random difference

is small.

Table 1: Complete ApET versus random pruning. ApET's advantage is large on TextVQA, small on GQA and undetectable on MMStar, and it comes mainly from fewer lost answers. Scores in %; random is mean ± sample SD over three seeds; CI is a paired 95% bootstrap interval over images; "Lost" counts baseline-correct answers that became wrong (TextVQA: full-credit correctness).

Dataset (n)	Schedule (avg.)	Uncompr.	ApET (share kept)	Random	ApET − random [95% CI]	Lost: ApET / rand.
TextVQA (100)	96→32 (64)	49.70	41.50 (83.5%)	20.63 ± 3.12	+20.87 [13.93, 29.03]	10 / 28.7 of 44
TextVQA (100)	288→96 (192)	49.70	44.20 (88.9%)	34.77 ± 4.80	+9.43 [3.20, 15.43]	7 / 16.0 of 44
GQA (800)	96→32 (64)	69.00	63.75 (92.4%)	61.29 ± 0.79	+2.46 [0.25, 4.71]	78 / 97.3 of 552
GQA (800)	288→96 (192)	69.00	68.00 (98.6%)	66.83 ± 0.62	+1.17 [−0.58, 2.83]	34 / 44.7 of 552
MMStar (1500)	96→32 (64)	34.73	33.40 (96.2%)	33.53 ± 0.53	−0.13 [−1.35, 1.07]	79 / 76.0 of 521
MMStar (1500)	288→96 (192)	34.73	35.00 (100.8%)	35.20 ± 0.29	−0.20 [−1.25, 0.85]	36 / 39.0 of 521

4.5 Finding 4: similar accuracy hides answer turnover

Relative to the uncompressed model, ApET changes correctness in both directions in all six conditions (gains at 64/192 tokens: TextVQA 3/3, GQA 36/26, MMStar 59/40). At GQA 288→96, 34 losses and 26 gains give only a 1-point net drop; at MMStar 288→96, 36 losses and 40 gains slightly raise accuracy (34.73%→35.00%; the interval includes zero). Accuracy alone therefore cannot establish answer preservation. Some gains may be scoring or format changes (e.g., "Steigl"→"Stiegl") rather than better visual evidence; we have not yet inspected them systematically.

4.6 Reconciling the findings

Finding 1 concerns individual token rankings; Finding 3 concerns the whole procedure, which differs from random pruning in selection, basis handling and merging at once. Two non-exclusive hypotheses remain open:

- **H1 (selection):** E-based selection keeps more useful tokens than random selection, e.g., by discarding easily reconstructed padding or background (low E on padding is visible in examples but not measured systematically). *Test:* vary only selection, with basis handling and merging fixed, on an unselected subset.
- **H2 (representation preservation):** preserved basis representations and merging keep information that random pruning drops. Finding 2 suggests merging gives retained tokens different roles. *Test:* merging on vs. off under matched selection.

The diagnostic set cannot separate them. It is also where the impression behind our random comparison arose (with basis and merging fixed, random retention kept 23–27 of 30 answers vs. 19–21 for E-based selection), but it was selected from ApET's own outcomes, so its 11 ApET losses are failures by construction; Table 1 corrects this impression at scale.

4.7 RQ3 feasibility, interpretation and validity

RQ3 feasibility. In an exploratory five-fold analysis on saved data, three image-level E features gave weak prediction of ApET losses (out-of-fold AUROC 0.41–0.57) and no clear Brier-score improvement over the base rate. This does not support the tested image-only adaptive budget; other policies remain possible. At 96→32, 56 of 100 GQA images had both lost and preserved questions under the same compressed input, motivating question-aware approaches.

- **Worked:** ApET clearly beats random pruning on TextVQA, mainly via fewer losses.
- **Did not work:** E does not consistently rank tokens by either measured response on the diagnostic set; image-level E features did not predict failures.
- **Surprising failure modes:** merging can corrupt a word ("Brolio"→"Bottoli"; undoing one retained token's merge restores it, unlike control restorations; one post-hoc case).
- **Goals not yet met:** component attribution (RQ2) and a supported RQ3 direction.

Validity checks. Target and scoring audits; full-576 identity (no-compression) controls and fixed-feature replays matched reference outputs exactly; all 90 recomputed correlations matched saved values; loss/gain counts satisfy correct = baseline − losses + gains in all 27 conditions. Limits: small, failure-enriched RQ1 set; first-token responses; two layers on six questions; development subsets (TextVQA: 100); intervals omit run-to-run variation; no timing.

5. Roadblocks, Risks and Mitigations

- **Integration:** changing Kaggle package defaults broke ApET/LLaVA compatibility. *Action:* locked environment with recorded fixes; resolved.
- **Run-to-run variation:** some fresh compressed runs change answers, although replays from saved features are exact. A targeted diagnostic found identical inputs, FPS selections and merge assignments, but post-merge features differing by up to 0.016. *Root-cause hypothesis:* nondeterministic FP16 accumulation in the CUDA `scatter_add_` used for merging (not yet isolated). *Action:* keep first runs as primaries; repeat critical cases.
- **Selection bias:** the diagnostic set was chosen from ApET's outcomes. *Action:* restrict it to RQ1 and case studies.

Table 2: Risk register: rows match the roadblocks above, plus one planning risk (likelihood/impact qualitative).

Risk	Likel./impact	Status	Contingency
Integration	Low / Med.	Green	Rebuild from the locked specification
Run-to-run variation	Med. / Med.	Yellow	Keep first runs as primaries; repeat critical cases
Selection bias	Med. / High	Yellow	Use unselected subsets for component comparisons
Attribution unclear; no RQ3 candidate	Med. / Med.	Yellow	Report a well-controlled negative result

6. Updated Plan and Milestones

- **Delta from proposal:** main comparisons ran in October, and the planned random baseline was run at scale first (Section 3); alternative bases are deprioritized (FPS and random bases perform similarly in [1]); RQ2 is now the main deliverable, RQ3 optional.
- **Next steps (explain first, then explore):** (1) *Separate the components* on an outcome-unselected cohort: E-based vs. random selection with basis handling and the merge rule fixed, and merging vs. dropping with selection fixed. (2) *Roles of retained tokens:* study merge loads and feature changes (e.g., how averaged tokens differ from the originals) alongside answer turnover, across stages and tasks, to test whether collecting discarded information helps some answers while disturbing others. (3) *RQ3, exploratory:* no fixed direction yet; evaluate one modest change only if RQ1/RQ2 evidence identifies a consistent problem.
- **Evaluation plan for final:** random pruning and fixed-budget ApET baselines; 96→32 primary, 288→96 if feasible.
- **Success criteria:** *Minimum:* one controlled comparison isolating a single ApET component or operation on $\geq$100 outcome-unselected questions (matched budgets, three random seeds where relevant), reporting accuracy, APR, losses/gains and paired 95% intervals; reproducible negative or inconclusive results count. *Stretch:* if supported, an RQ3 change compared with fixed-budget ApET at equal average tokens on held-out questions. Significance is reported, not required.

Table 3: Milestones.

Target	Deliverable and completion criterion
Oct 30	Submit report PDF and ∼five-slide deck
Nov 6	Choose one comparison and write its protocol (conditions, subsets, analysis)
Nov 13	Implement it and validate on a small unselected pilot (identity and replay checks pass)
Nov 20	Run the main comparison at 96→32 (288→96 if feasible) and analyze it
Nov 24	RQ3 go/no-go; optionally a small test of one modification
Dec 15 (tentative)	Final presentation and report/code submission

7. References

[1] Q. Ma et al. ApET: Approximation-Error Guided Token Compression for Efficient VLMs. CVPR, 2026.
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